# Analysis and Evaluation of Machine Learning Operations on a Show Recommendation System

Marco De Rito – Data Engineer

Università degli Studi di Trieste

September 3, 2024

# Introduction

**Objective:** Develop a recommendation system for a streaming platform using machine learning techniques.

- Personalized user profiles based on viewing habits.
- Segmenting the catalogue into distinct clusters.
- Enhance user satisfaction by providing relevant content.

# Data and Preprocessing

**Datasets:**

- *Netflix Movies and TV Shows* (Kaggle)
- User ratings integrated from additional sources.

**Preprocessing Steps:**

- Handling missing data.
- Textual data transformation using TF-IDF:

$$\text{TF-IDF}(t, d) = \text{TF}(t, d) \times \text{IDF}(t)$$

# Visualization of Data and Results

**Genre Distribution:**

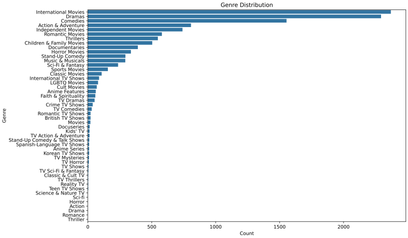


Figure: Genre Distribution

# Visualization of Data and Results

## Pairplot of Numerical Variables:

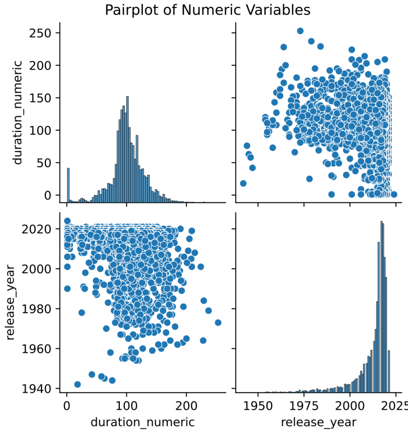


Figure: Pairplot of Numerical Variables

# Visualization of Data and Results

**Like/Dislike Distribution:**

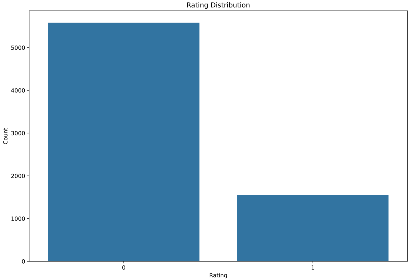


Figure: Like/Dislike Distribution

# Analysis of Statistical Results

**Model Accuracy:** 82%

- **Confusion Matrix:**
- **Classification Report:**
- **ROC Curve:**
- **Precision-Recall Curve:**
- **Feature Importance:**

# Model Training and Testing

**Data Splitting:**

- 80% Training, 20% Testing

**Class Imbalance Handling:**

- SMOTE for oversampling minority class.

# Clustering Using K-Means

**K-Means Algorithm:**

- Partitioning data into $k$ clusters.
- Optimization based on Euclidean distance.
- Silhouette Score for determining optimal $k$:

$$\text{Silhouette Score} = \frac{b - a}{\max(a, b)}$$

# Random Forest Classifier for Recommendation

**Random Forest Overview:**

- Ensemble method using decision trees.
- Voting mechanism for final output.
- Robustness against overfitting.

# Evaluation Metrics: Confusion Matrix

**Confusion Matrix:**

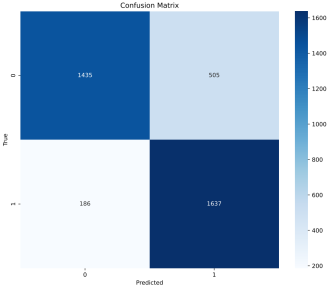


Figure: Confusion Matrix

**Explanation:** The confusion matrix provides a summary of prediction results on the classification problem. The number of correct and incorrect predictions are summarized with count values and broken down by each class.

# Evaluation Metrics: Classification Report

**Classification Report:**

- Precision $(P)$: $P = \frac{TP}{TP+FP}$
- Recall $(R)$: $R = \frac{TP}{TP+FN}$
- F1-Score: $F1 = 2 \cdot \frac{P \cdot R}{P+R}$

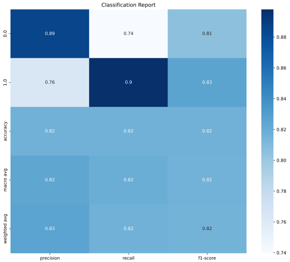


Figure: Classification Report

**Explanation:** This report shows the main classification metrics: precision, recall, and F1-score for each class, helping to assess the performance of the classifier.

# Evaluation Metrics: ROC Curve and AUC

**ROC Curve and AUC:**

- ROC Curve: Plot of True Positive Rate (TPR) vs. False Positive Rate (FPR).
- AUC (Area Under the Curve): Measures overall performance. AUC closer to 1 indicates better performance.

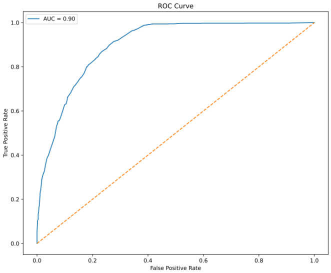


Figure: ROC Curve with AUC = 0.90

**Explanation:** The ROC curve is a graphical plot that illustrates the diagnostic ability of a binary classifier. The AUC is a single number summary of the model performance.

# Evaluation Metrics: Precision-Recall Curve

**Precision-Recall Curve:**

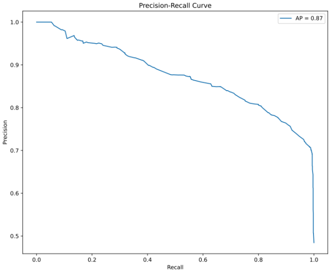


Figure: Precision-Recall Curve with AP $= 0.87$

**Explanation:** The precision-recall curve is a useful tool for understanding the tradeoff between precision and recall for different threshold settings of the classifier.

# Evaluation Metrics: Feature Importance

**Feature Importance:**

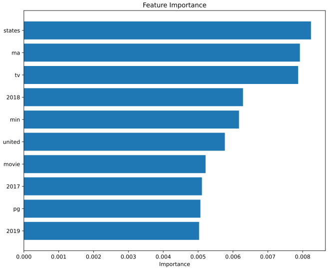


Figure: Feature Importance Analysis

**Explanation:** Feature importance scores highlight which features are most influential in predicting the target variable. This helps in understanding the model's decision process.

# Discussion and Conclusion

**Key Takeaways:**

- K-Means and Random Forest effectively segment and recommend.
- High accuracy with good generalization.

**Limitations:**

- Potential "filter bubble" effect.